

Motivation of Mixture of Experts

- As model size increases, **activating all parameters** for each inference step becomes computationally expensive.
- Not all model parameters* are useful in the forward pass. We **activate only a small subset of parameters**, thereby reducing computational cost.

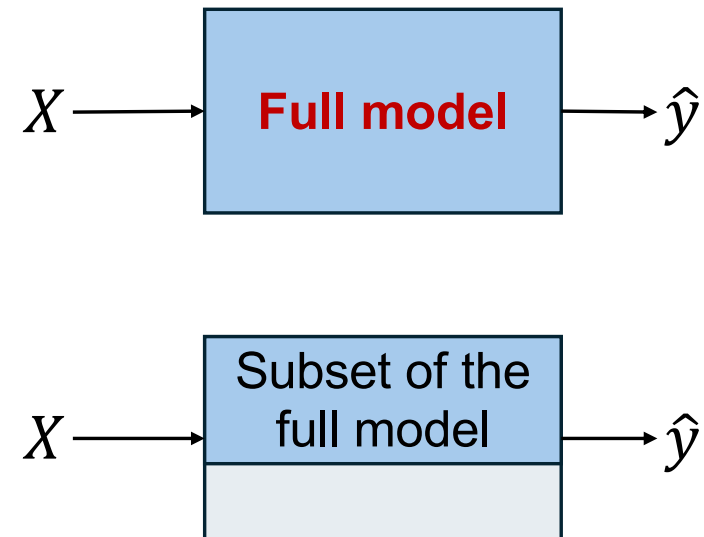
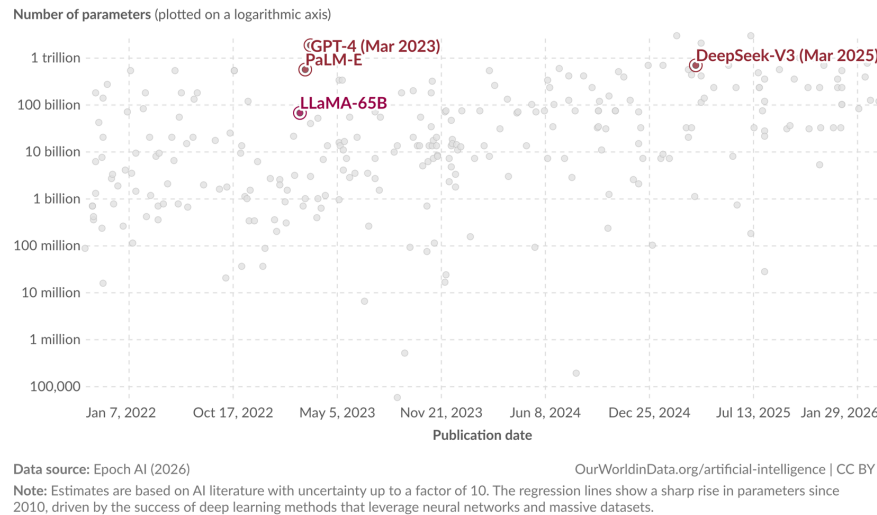
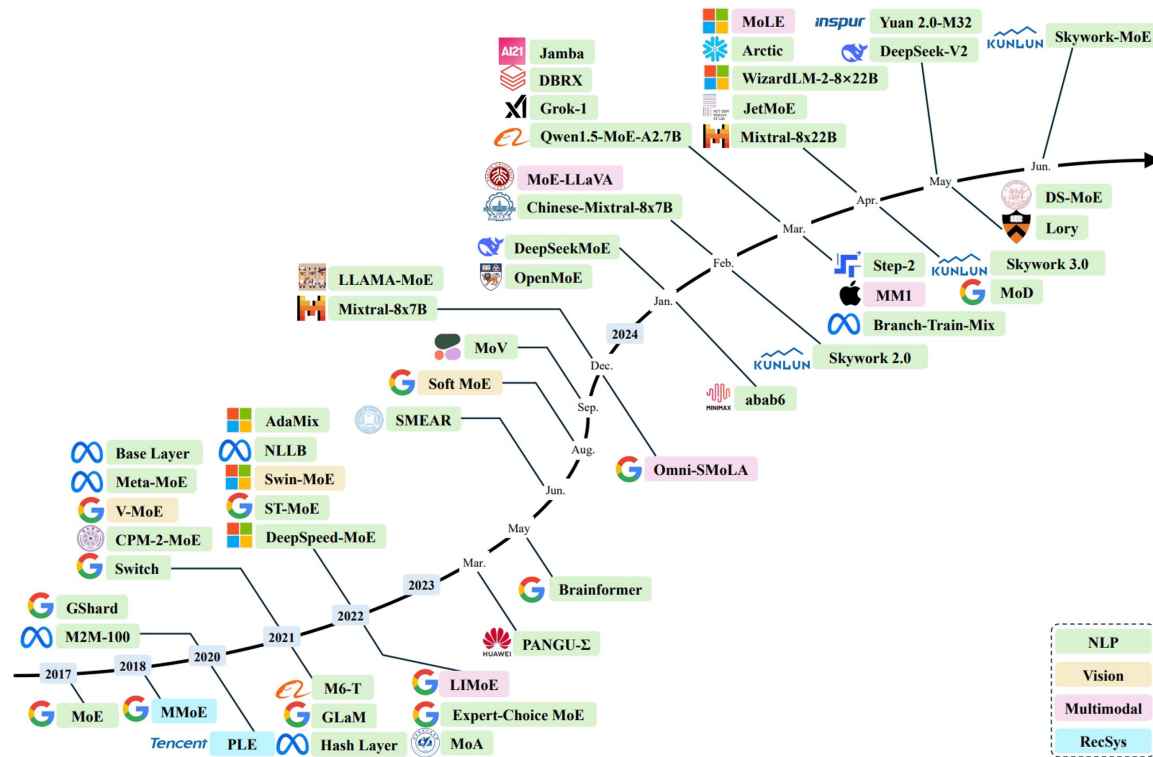


Image source: <https://ourworldindata.org/grapher/exponential-growth-of-parameters-in-notable-ai-systems>

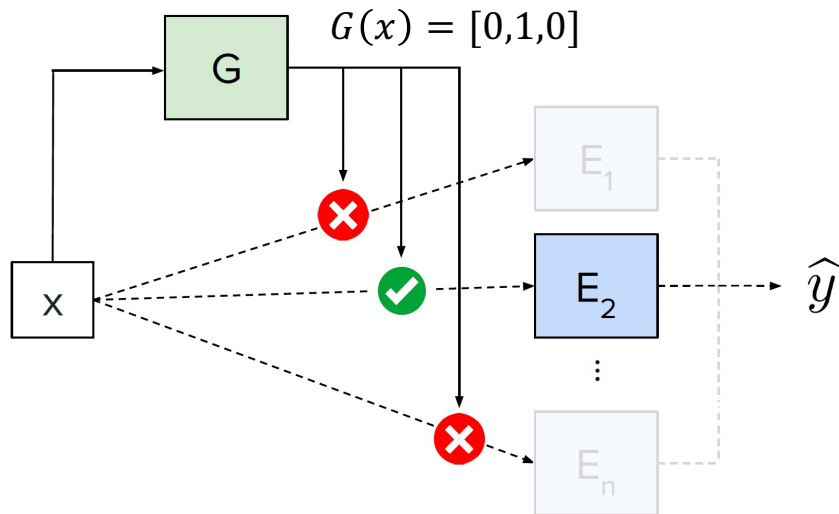
Road map of Mixture of Experts



- This figure summarizes the evolution of MoE models from 2017 to 2024.
- MoE has expanded from *an early research idea* to a widely used architecture in **NLP, vision, multimodal, and recommender systems**.

Cai W, Jiang J, Wang F, et al. A survey on mixture of experts in large language models[J]. IEEE Transactions on Knowledge and Data Engineering, 2025.

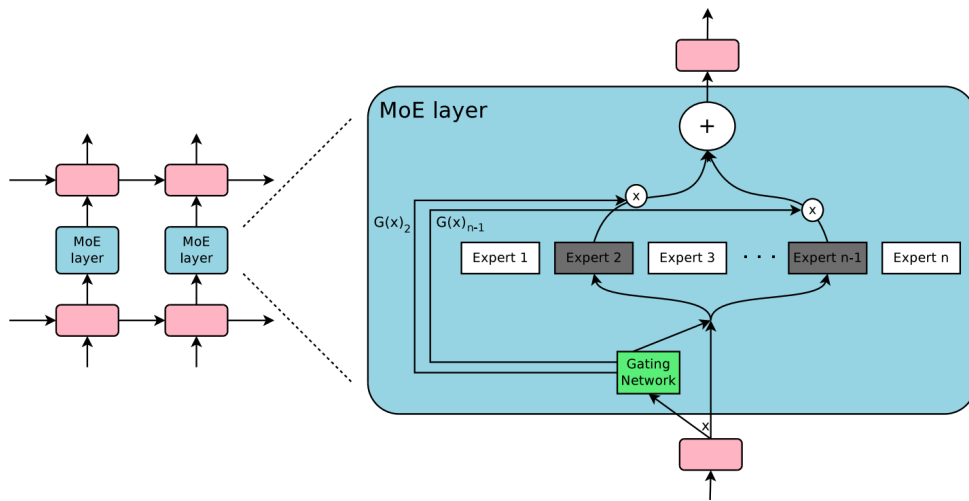
Overview of Mixture of Experts



- Input x is sent to a **gating network** G
- The gating network decides which expert is the most suitable for the current input (using a one-hot vector).
- Instead of using all experts $E_1, E_2, \dots, E_n$, the model activates only a small subset.
- In the example on the left, only E_2 is selected, while the other experts remain inactive.
- The final output is generated from the selected expert:

$$\hat{y} = \sum_{i \in \mathcal{I}_k} G(x)_i E_i(x)$$

Mixture of Experts in LSTM



- For input X , the gating network computes routing scores: $G = \text{Softmax}(W_G X)$
- The model selects the top- K experts, e.g. $K=2$, $I = \{i_0, i_1\} = \text{TopK}(G, k = 2)$
- The selected experts are assigned weights:

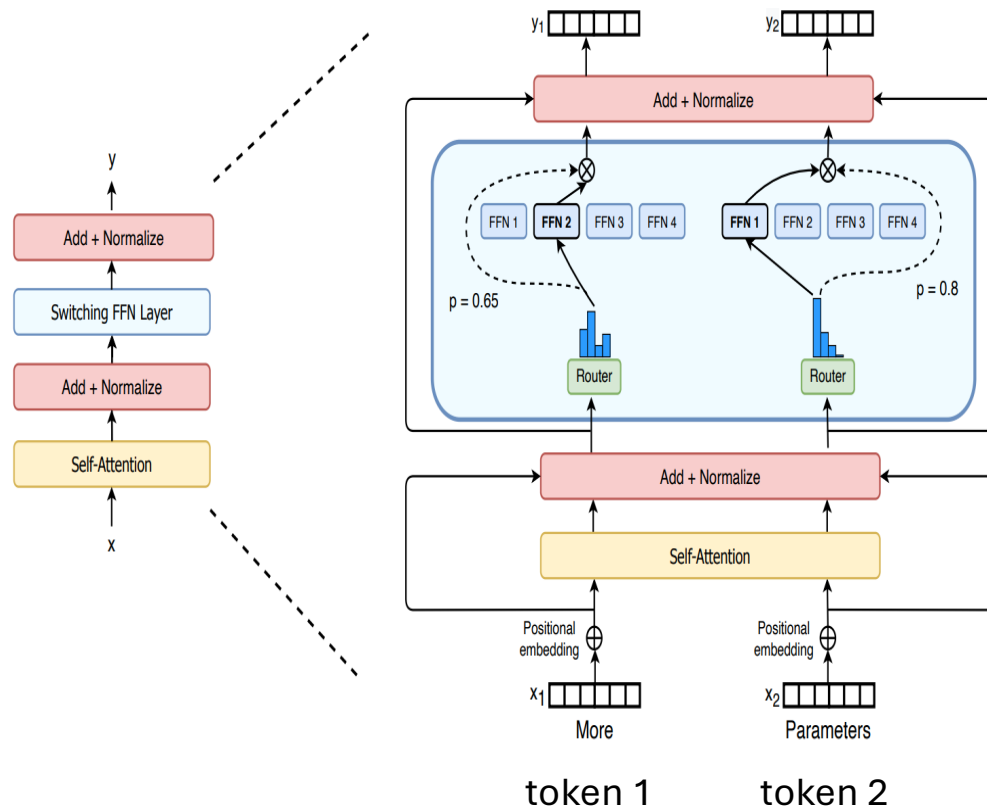
$$s_0 = \frac{G_{i_0}}{(G_{i_0} + G_{i_1})}, s_1 = \frac{G_{i_1}}{(G_{i_0} + G_{i_1})}$$

- Each selected expert is a **feed-forward network**. The final output is a weighted combination of the selected experts:

$$Y = s_0 \text{FFN}_{i_0}(X) + s_1 \text{FFN}_{i_1}(X)$$

Shazeer N, Mirhoseini A, Maziarz K, et al. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. arXiv preprint arXiv:1701.06538, 2017.

Mixture of Experts in Transformer

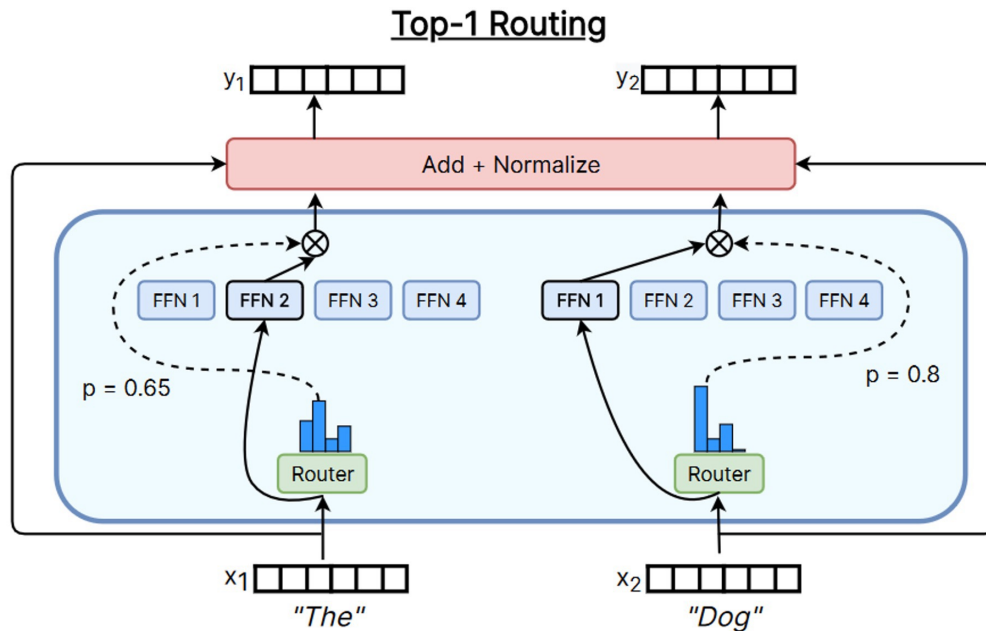


- Switch Transformer block, where the **standard FFN** is replaced by a **Mixture-of-Experts layer**.
- For each input token, the router computes a probability: $P = \text{Softmax}(W_r h)$
- Each expert is a feed-forward network E_i . The model selects the expert with the **highest routing probability**
- Only the selected expert processes the token:

$$\text{MoE}(h) = p_k E_k(h)$$

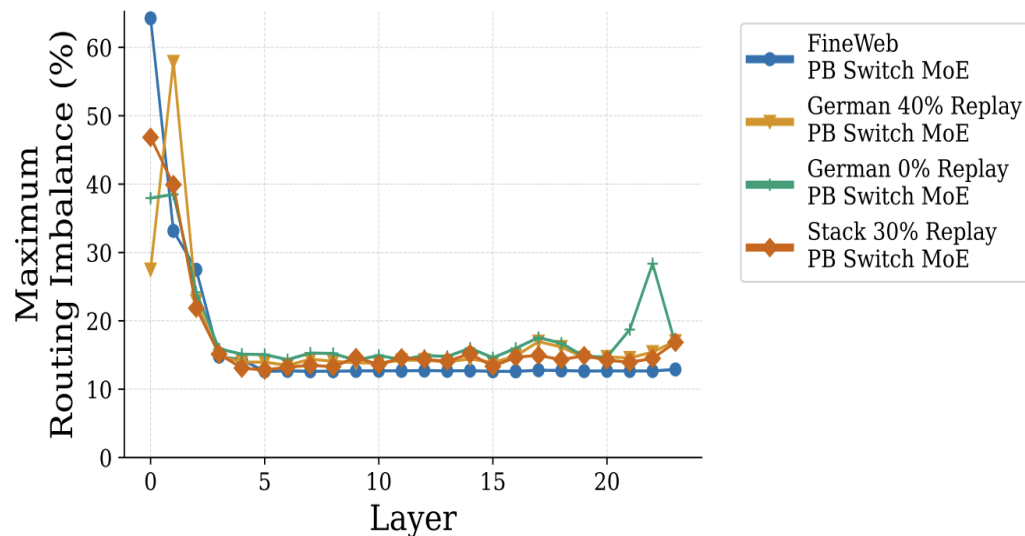
Fedus W, Zoph B, Shazeer N. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity[J]. Journal of Machine Learning Research, 2022, 23(120): 1-39.

Routing algorithms — Top 1 routing



- **Advantages:** each token activates only **one expert**, minimizing forward-pass FLOPs.
- **Limitations: routing collapse.** During training, the gating network may over-select a small subset of experts, leaving others rarely used and therefore under-trained.

Routing algorithms — Routing collapse

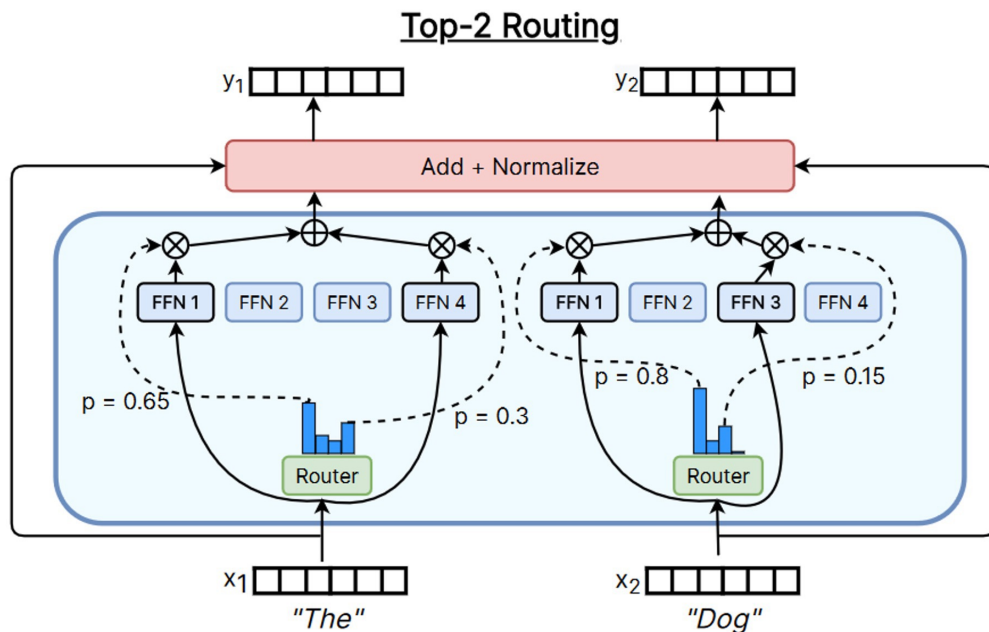


(a) PBTk Switch MoE, FineWeb

- Maximum Routing Imbalance (%) measures how unevenly tokens are distributed across experts within a layer.
- A **higher value** means that routing is more concentrated on a **small** number of experts, indicating poorer load balance
- MoE routers may overuse a few experts while underutilizing others in early layers

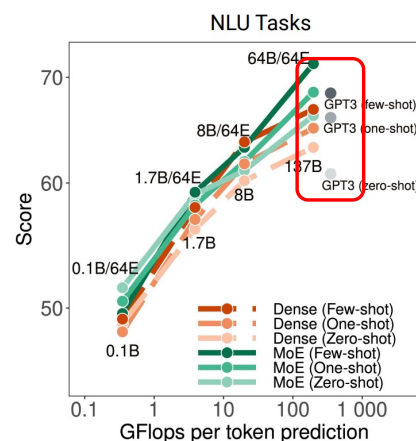
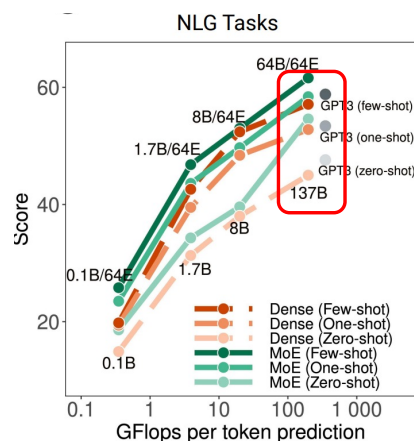
Thérien B, Joseph C É, Sarwar Z, et al. Continual Pre-training of MoEs: How robust is your router?. arXiv preprint arXiv:2503.05029, 2025.

Routing algorithms — Top 2 routing



- **Advantages:** this improves representation power and can alleviate routing collapse
- **Limitations:** Each token activates two experts, so the computational cost is higher. Communication overhead is also larger, since more tokens must be transferred across devices.

Case of Mixture of Experts



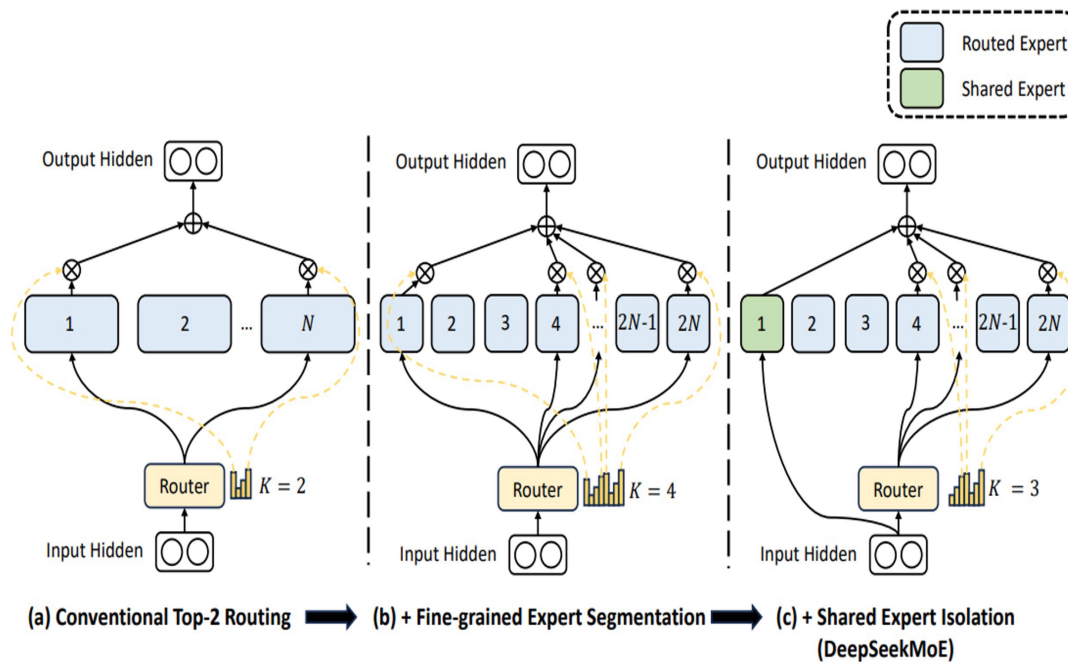
- Using similar FLOPs per token prediction, MoE models have better performance than the dense variants.

		GPT-3	GLaM	relative
cost	FLOPs / token (G)	350	180	-48.6%
	Train energy (MWh)	1287	456	-64.6%
accuracy on average	Zero-shot	56.9	62.7	+10.2%
	One-shot	61.6	65.5	+6.3%
	Few-shot	65.2	68.1	+4.4%

- GLaM has better performance while using **1/3** of the energy and **1/2** of serving cost of GPT-3.

Glam: Efficient scaling of language models with mixture-of-experts. International conference on machine learning, 2022

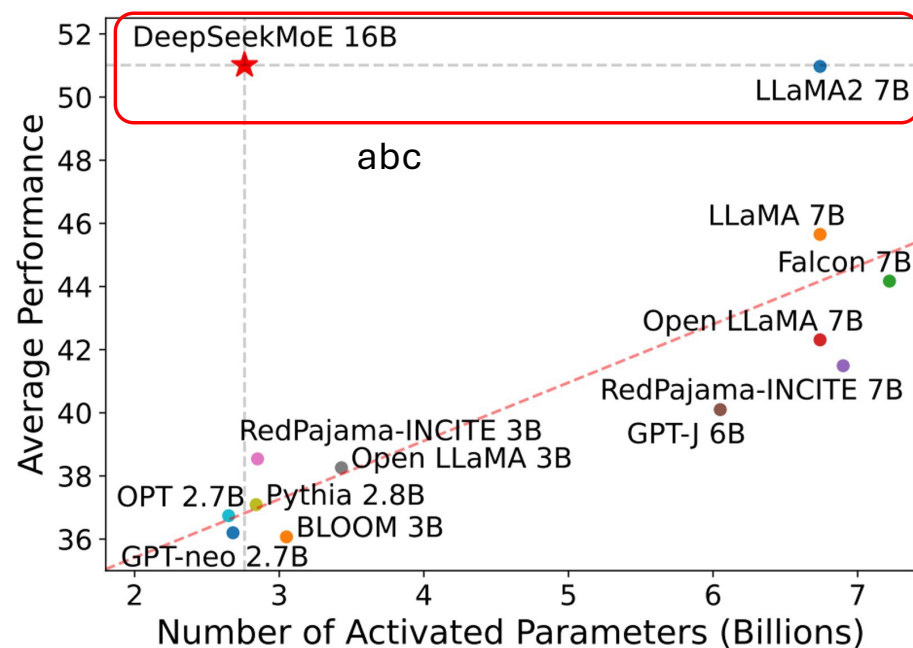
Case of Mixture of Experts--DeepSeek



- Each MoE layer consists of 2 shared experts and 64 routed experts (select 6 experts)
- **Shared experts:** each token passes through (no routing). Shared experts **learn general knowledge**, such as syntax and common semantics.
- **Routed experts:** selectively activated for each token by a router. Routed experts focus more on **domain-specific knowledge**

Deepseekmoe: Towards ultimate expert specialization in mixture-of-experts language models.ACL. 2024

Case of Mixture of Experts—DeepSeek



DeepSeek-MoE-16B, matches or even surpasses LLaMA2-7B, which activates all 7B parameters

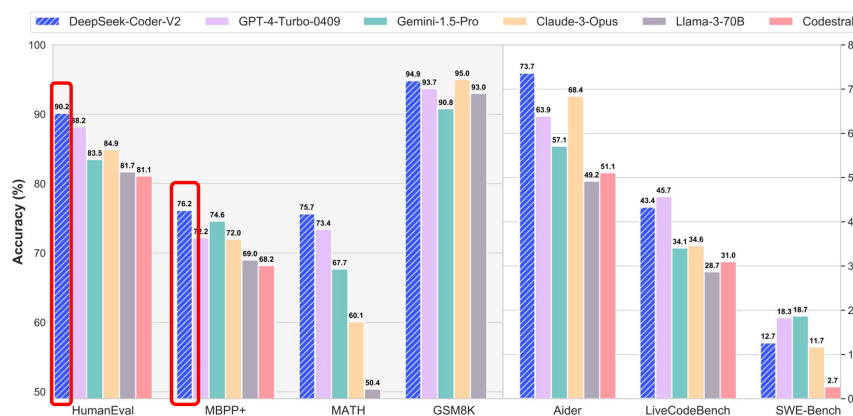
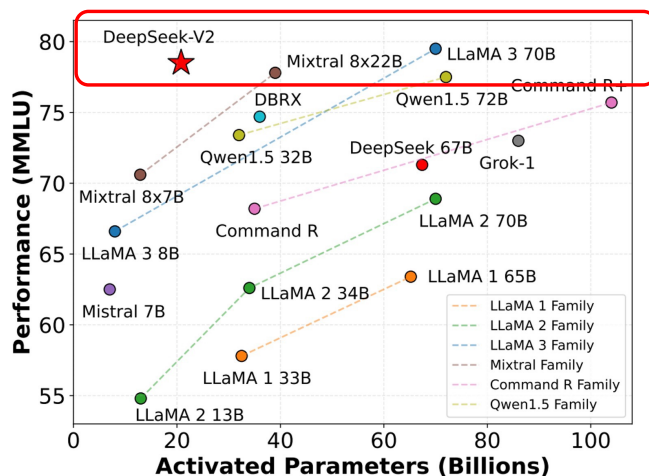
Metric	# Shot	DeepSeek 7B (Dense)	DeepSeekMoE 16B
# Total Params	N/A	6.9B	16.4B
# Activated Params	N/A	6.9B	2.8B
FLOPs per 4K Tokens	N/A	183.5T	74.4T
# Training Tokens	N/A	2T	2T
Pile (BPPB)	N/A	0.75	0.74
HellaSwag (Acc.)	0-shot	75.4	77.1
PIQA (Acc.)	0-shot	79.2	80.2
ARC-easy (Acc.)	0-shot	67.9	68.1
ARC-challenge (Acc.)	0-shot	48.1	49.8
RACE-middle (Acc.)	5-shot	63.2	61.9
RACE-high (Acc.)	5-shot	46.5	46.4
DROP (EM)	1-shot	34.9	32.9
GSM8K (EM)	8-shot	17.4	18.8
MATH (EM)	4-shot	3.3	4.3
HumanEval (Pass@1)	0-shot	26.2	26.8
MBPP (Pass@1)	3-shot	39.0	39.2
TriviaQA (EM)	5-shot	59.7	64.8
NaturalQuestions (EM)	5-shot	22.2	25.5
MMLU (Acc.)	5-shot	48.2	45.0
WinoGrande (Acc.)	0-shot	70.5	70.2
CLUEWSC (EM)	5-shot	73.1	72.1
CEval (Acc.)	5-shot	45.0	40.6
CMMLU (Acc.)	5-shot	47.2	42.5
CHID (Acc.)	0-shot	89.3	89.4

DeepSeek-MoE-16B, with only 2.8B activated parameters

A few cases where MoE is worse than the dense model.

Deepseekmoe: Towards ultimate expert specialization in mixture-of-experts language models.ACL. 2024

Case of Mixture of Experts—DeepSeek



Model	API Price / 1M Tokens	
	Input \$	Output \$
DeepSeek-Coder-V2	0.14	0.28
DeepSeek-V2	0.14	0.28
GPT-4-Turbo-1106	10.00	30.00
GPT-4-0613	30.00	60.00
GPT-3.5	1.50	2.00
Gemini 1.5 Pro	7.00	21.00
Claude 3 Opus	15.00	75.00
Claude 3 Sonnet	3.00	15.00
Claude 3 Haiku	0.25	1.25
abab-6.5 (MiniMax)	4.14	4.14
abab-6.5s (MiniMax)	1.38	1.38
ERNIE-4.0 (文心一言)	16.56	16.56
GLM-4 (智谱清言)	13.80	13.80
Moonshot-v1 (月之暗面)	3.32	3.32
Qwen1.5 72B (通义千问)	2.76	2.76
LLaMA 3 70B	3.78	11.34
Mixtral 8x22B	2.00	6.00

Deepseek-v2

236B total parameters, 21B are activated.

It achieves strong performance with much lower activated computation

Deepseek-Coder-V2

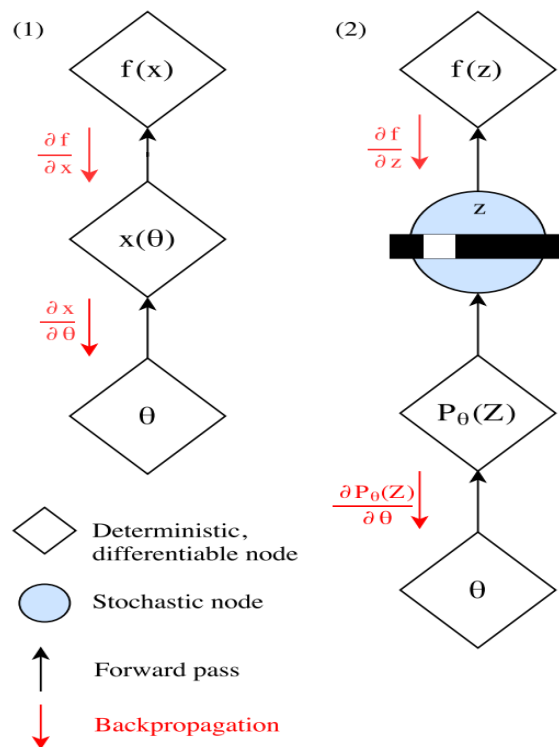
Continue pretraining from an intermediate checkpoint of Deepseek-V2 on 6T, adding to a total of 10.2T pre-training tokens. Comparable to GPT4-Turbo in code-specific tasks.

Deepseek-Coder-V2

Reduced price per 1M tokens

Liu A, Feng B, Wang B, et al. Deepseek-v2: A strong, economical, and efficient mixture-of-experts language model. arXiv preprint arXiv:2405.04434, 2024.

MoE training: backprop through expert selection



- Panel (1): ordinary neural network training. In a standard dense model, every operation is differentiable
- Panel (2): MoE routing creates a stochastic or discrete node. The selection of experts step is like the discrete node z in the figure. The gradients cannot pass through that choice in the normal way.

Jang E, Gu S, Poole B. Categorical reparameterization with gumbel-softmax. arXiv preprint arXiv:1611.01144, 2016.

MoE training

Given a token sequence of length T , $(x_1, x_2, \dots, x_T)$, the training loss can be written as:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{aux}}$$

$$\mathcal{L}_{\text{task}} = -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(x_t | x_{<t})$$

$$\mathcal{L}_{\text{aux}} = \alpha N \sum_{i=1}^N f_i P_i$$

$$\begin{aligned} \mathbf{h}_t &= \mathbf{u}_t + \sum_{i=1}^N g_{i,t} \text{FFN}_i(\mathbf{u}_t), \\ g_{i,t} &= \begin{cases} s_{i,t}, & s_{i,t} \in \text{Topk}(\{s_{j,t} \mid 1 \leq j \leq N\}, K), \\ 0, & \text{otherwise,} \end{cases} \\ s_{i,t} &= G(\mathbf{u}_t^T \mathbf{e}_i), \end{aligned}$$

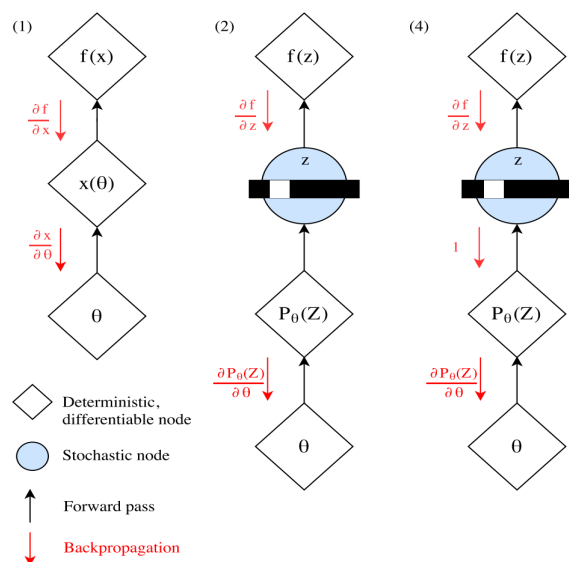
Non-differentiable

$$\begin{aligned} f_i &= \frac{N}{KT} \sum_{t=1}^T \mathbb{1}(\text{Token } t \text{ selects Expert } i), \\ P_i &= \frac{1}{T} \sum_{t=1}^T s_{i,t}, \end{aligned}$$

- $\mathcal{L}_{\text{task}}$ is the cross-entropy loss for next-token prediction
- $\mathcal{L}_{\text{aux}}$ encourages balanced expert utilization (details in the next page).

Jang E, Gu S, Poole B. Categorical reparameterization with gumbel-softmax. arXiv preprint arXiv:1611.01144, 2016.

MoE training



Panel (4): In the forward pass, the model still uses a hard or sampled value z . In the backward pass, it **approximates the gradient**.

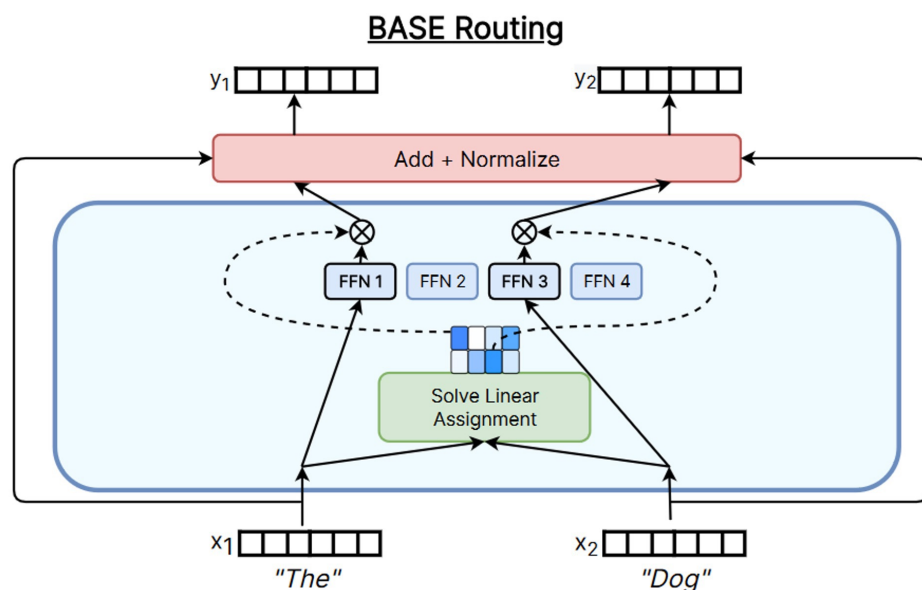
For the auxiliary loss:

$$\mathcal{L}_{\text{aux}} = \alpha N \sum_{i=1}^N f_i P_i$$

- f_i is the fraction of tokens actually routed to expert i , so it depends on a hard routing decision and is **not differentiable**
- P_i is the average router probability for expert i , so it is **differentiable**
- As a result, the gradient mainly flows through P_i , while f_i is treated as a constant

Jang E, Gu S, Poole B. Categorical reparameterization with gumbel-softmax. arXiv preprint arXiv:1611.01144, 2016.

Routing algorithms — BASE routing



- **Advantages:** treat token-to-expert routing as a linear assignment problem. It provides **strict load balancing** across experts
- **Limitations:** BASE routing **unsuitable** for autoregressive decoding, where tokens are generated one by one. Solving the assignment problem is computationally expensive.

Lewis M, Bhosale S, Dettmers T, et al. Base layers: Simplifying training of large, sparse models. International Conference on Machine Learning. PMLR, 2021: 6265-6274.

Routing algorithms — BASE routing

Linear assignment problem

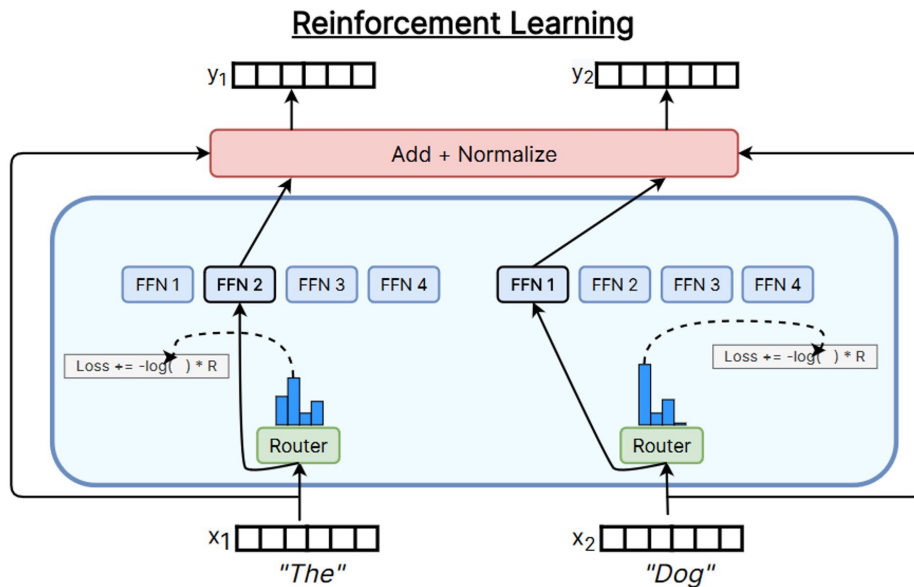
$$\begin{aligned} & \text{maximize } \sum_t h_t \cdot w_{a_t} \\ & \text{subject to } \forall e \sum_{t=0}^T \mathbb{1}_{a_t=e} = \frac{T}{E} \end{aligned}$$

- This problem maximizes the sum of token–expert matching scores
- The constraint requires every expert to receive exactly T/E tokens. Therefore, all experts **handle the same number of tokens**

- h_t : the representation of token t
- w_{a_t} : the parameter of expert assigned to token t
- a_t : the expert selected for token t
- T : the total number of tokens in the batch
- E : the number of experts
- $\mathbb{1}_{a_t=e}$: an indicator function, equal to 1 if token t is assigned to expert e , and 0 otherwise.

Lewis M, Bhosale S, Dettmers T, et al. Base layers: Simplifying training of large, sparse models. International Conference on Machine Learning. PMLR, 2021: 6265-6274.

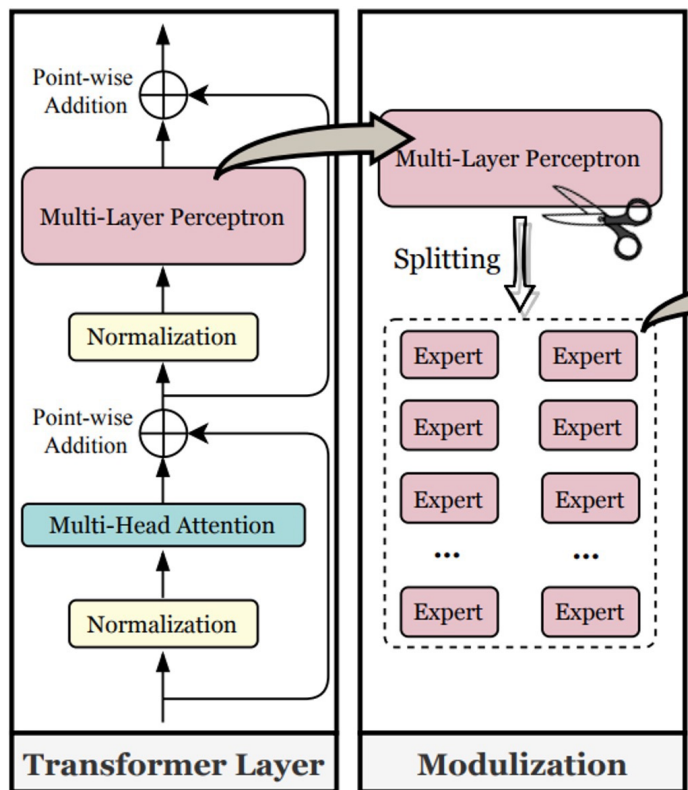
Routing algorithms — Reinforcement learning



- In Top-k routing, expert selection is a discrete decision. In practice, training usually updates *only the softmax gate values*, not the discrete expert assignment itself.
- RL-based routing addresses this by treating routing as a policy learning problem:
 - the router is the agent
 - expert selection is the action
 - task performance is the reward
- **Limitations:** the reward signal is often sparse and training usually has high variance

Zuo S, Liu X, Jiao J, et al. Taming sparsely activated transformer with stochastic experts. arXiv preprint arXiv:2110.04260, 2021.

Building MoE LLMs



Sparse transformation

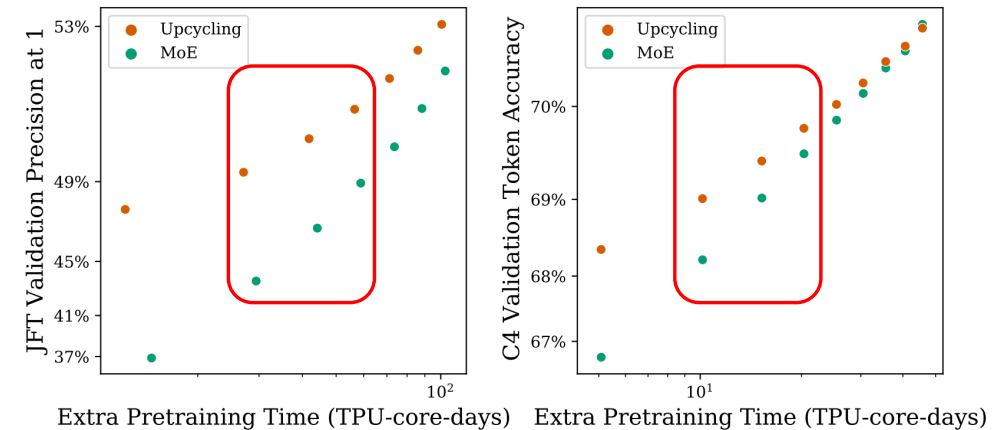
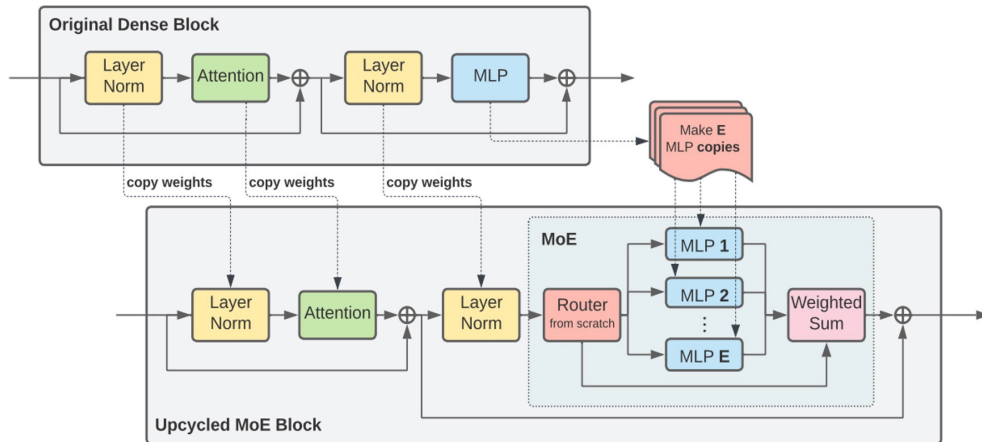
Sparse Upcycling

Copy FFNs to form experts → Parameters increased

Split FFNs to form experts → Parameters unchanged

Sparse Splitting

Building MoE LLMs—Sparse Upcycling



- **Advantages:** it can reuse the pretrained knowledge of an dense model.
- **Limitations:** the total number of parameters increases by a factor of N, leading to higher storage and communication costs

- To build a Mixture-of-Experts (MoE) model, "**recycling**" a pretrained dense model is more **economical and effective** than starting from zero.

Komatsuzaki A, Puigcerver J, Lee-Thorp J, et al. Sparse upcycling: Training mixture-of-experts from dense checkpoints. arXiv preprint arXiv:2212.05055, 2022.

Sparse Upcycling-Mixtral MoE

Model	Active parameters	Common sense and reasoning					Knowledge	
		MMLU	HellaS	WinoG	Arc C (5)	Arc C (25)	TriQA	NaturalQS
LLaMA 2 70B	70B	69.9%	87.1%	83.2%	86.0%	85.1%	77.57%	35.5%
Command R	35B	68.2%	87.0%	81.5%	-	66.5%	-	-
Command R+	104B	75.7%	88.6%	85.4%	-	71.0%	-	-
Mistral 7B	7B	62.47%	83.1%	78.0%	77.2%	78.1%	68.8%	28.1%
Mixtral 8x7B	12.9B	70.63%	86.6%	81.2%	85.8%	85.9%	78.4%	36.5%
Mixtral 8x22B	39B	77.75%	88.5%	84.7%	91.3%	91.3%	82.2%	40.1%

- **Mixtral 8x22B** (only 39B active parameters) outperforms much larger dense models like Command R+ (104B) and LLaMA 2 70B across nearly all benchmarks.

Model	Active Params	MMLU	HellaS	WinoG	PIQA	Arc-e	Arc-c	NQ	TriQA	HumanE	MBPP	Math	GSM8K
LLaMA 2 7B	7B	44.4%	77.1%	69.5%	77.9%	68.7%	43.2%	17.5%	56.6%	11.6%	26.1%	3.9%	16.0%
LLaMA 2 13B	13B	55.6%	80.7%	72.9%	80.8%	75.2%	48.8%	16.7%	64.0%	18.9%	35.4%	6.0%	34.3%
LLaMA 1 33B	33B	56.8%	83.7%	76.2%	82.2%	79.6%	54.4%	24.1%	68.5%	25.0%	40.9%	8.4%	44.1%
LLaMA 2 70B	70B	69.9%	85.4%	80.4%	82.6%	79.9%	56.5%	25.4%	73.0%	29.3%	49.8%	13.8%	69.6%
Mistral 7B	7B	62.5%	81.0%	74.2%	82.2%	80.5%	54.9%	23.2%	62.5%	26.2%	50.2%	12.7%	50.0%
Mixtral 8x7B	13B	70.6%	84.4%	77.2%	83.6%	83.1%	59.7%	30.6%	71.5%	40.2%	60.7%	28.4%	74.4%

Table 2: Comparison of Mixtral with Llama. Mixtral outperforms or matches Llama 2 70B performance on almost all popular benchmarks while using 5x fewer active parameters during inference.

Jiang A Q, Sablayrolles A, Roux A, et al. Mixtral of experts. arXiv preprint arXiv:2401.04088, 2024.

- **Mixtral 8x7B** achieves stronger results than LLaMA 2 70B on several tasks

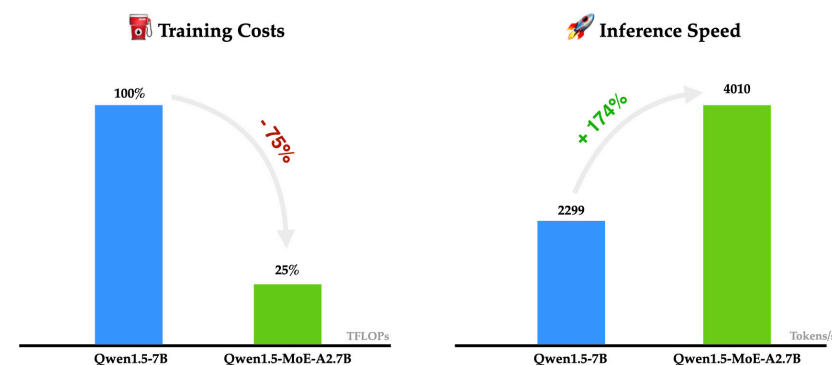
Sparse Upcycling-Qwen MoE

Model	MMLU	GSM8K	HumanEval	Multilingual	MT-Bench
Mistral-7B	64.1	47.5	27.4	40.0	7.60
Gemma-7B	64.6	50.9	32.3	-	-
Qwen1.5-7B	61.0	62.5	36.0	45.2	7.60
DeepSeekMoE 16B	45.0	18.8	26.8	-	6.93
Qwen1.5-MoE-A2.7B	62.5	61.5	34.2	40.8	7.17

*A2.7B=2.7B active parameters

- Upcycled from Qwen-1.8B, 14.3B parameters in total and 2.7B activated parameters
- It use shared 4 experts and routing experts 60 experts, choose 4

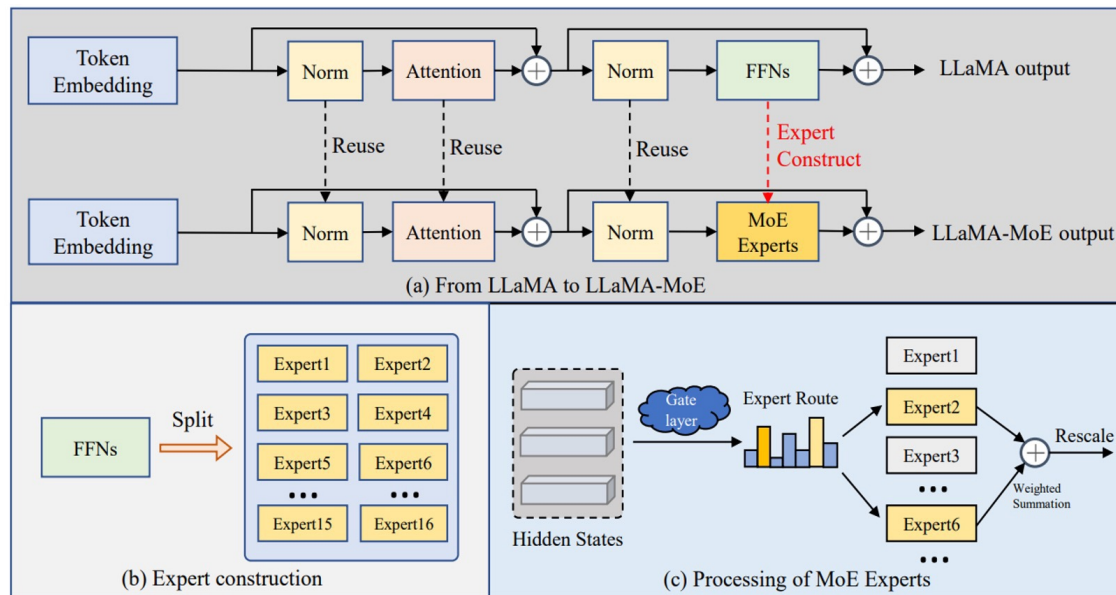
<https://qwen.ai/blog?id=qwen-moe>



A remarkable reduction of 75% in training

Inference speed up by 174%

Building MoE LLMs—Sparse Splitting



- **Advantages:** the total number of parameters remains unchanged, while the number of activated parameters during inference is reduced, leading to higher computational efficiency.
- **Limitations:** the expressive capacity of each expert is limited, as each expert has only $1/N$ of the original FFN capacity

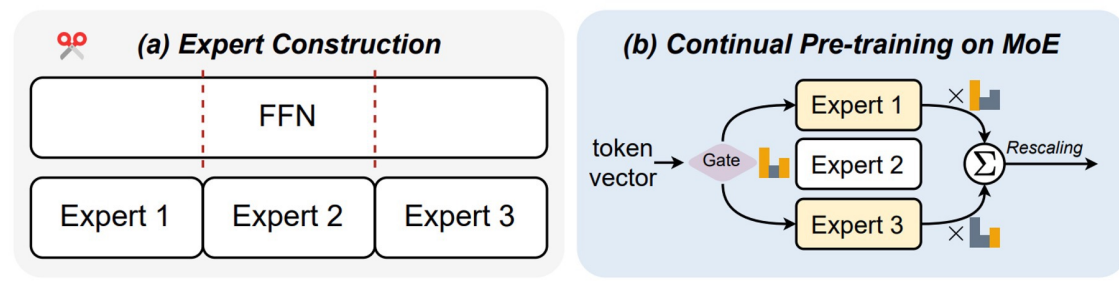
Zhu T, Qu X, Dong D, et al. Llama-moe: Building mixture-of-experts from llama with continual pre-training. Proceedings of the 2024 conference on empirical methods in natural language processing. 2024: 15913-15923.

Building MoE LLMs—Sparse Splitting

models of similar scale

Start from LLaMA-2-7B

Split into multiple experts



Baselines with similar number of active params

Model	#Activated Experts	#Total Experts	#Activated Params
OPT-2.7B	-	-	2.7B
Pythia-2.8B	-	-	2.8B
INCITE-Base-3B	-	-	2.8B
Open-LLaMA-3B-v2	-	-	3.4B
Sheared-LLaMA-2.7B	-	-	2.7B
LLaMA-MoE-3.0B (2/16)	2	16	3.0B
LLaMA-MoE-3.5B (4/16)	4	16	3.5B
LLaMA-MoE-3.5B (2/8)	2	8	3.5B

Commonsense & Reading Comprehension						
Model	SciQ	PIQA	WinoGrande	ARC-E	ARC-C (25)	HellaSwag (10)
OPT-2.7B	78.9	74.8	60.8	54.4	34.0	61.4
Pythia-2.8B	83.2	73.6	59.6	58.8	36.7	60.7
INCITE-Base-3B	85.6	73.9	63.5	61.7	40.3	64.7
Open-LLaMA-3B-v2	88.0	77.9	63.1	63.3	40.1	71.4
Sheared-LLaMA-2.7B	87.5	76.9	65.0	63.3	41.6	71.0
LLaMA-MoE-3.0B	84.2	77.5	63.6	60.2	40.9	70.8
LLaMA-MoE-3.5B (4/16)	87.6	77.9	65.5	65.6	44.2	73.3
LLaMA-MoE-3.5B (2/8)	88.4	77.6	66.7	65.3	43.1	73.3
Continued LM World Knowledge						
Model	LogiQA	BoolQ (32)	LAMBADA	NQ (32)	MMLU (5)	Average
OPT-2.7B	25.8	63.3	63.6	10.7	25.8	50.3
Pythia-2.8B	28.1	65.9	64.6	8.7	26.8	51.5
INCITE-Base-3B	27.5	65.8	65.4	15.2	27.2	53.7
Open-LLaMA-3B-v2	28.1	69.2	67.4	16.0	26.8	55.6
Sheared-LLaMA-2.7B	28.3	73.6	68.3	17.6	27.3	56.4
LLaMA-MoE-3.0B	30.6	71.9	66.6	17.0	26.8	55.5
LLaMA-MoE-3.5B (4/16)	29.7	75.0	69.5	20.3	26.8	57.7
LLaMA-MoE-3.5B (2/8)	29.6	73.9	69.4	19.8	27.0	57.6

Zhu T, Qu X, Dong D, et al. Llama-moe: Building mixture-of-experts from llama with continual pre-training. EMNLP. 2024: 15913-15923.

Quiz

1. Pen and paper; no compute or cellphone allowed.
 2. Turn in your answer sheet when you leave the classroom.
- Q1 (T/F): creating multiple experts from an existing LLM can be done by copy the original feed-forward network for several times.
 - Q2 (T/F): an MoE can select only 1 out of multiple experts per token.
 - Q3 (T/F): if an expert is not selected for generating a token w_t during training, that expert won't be used in generating future tokens starting from w_t
 - A: T; F; F